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Himani Sinhmar

Postdoctoral Research Associate, Princeton University

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Positions Held

Postdoctoral Research Associate, Leonard Lab — Sept. 2024 to present
Princeton University
Advisor: Prof. Naomi Ehrich Leonard

Education

Ph.D., Aerospace Engineering — Aug. 2019 to Aug. 2024
Cornell University
Specialization in Robotics; Minor in Computer Science
Doctoral Advisor: Prof. Hadas Kress-Gazit

B.Tech. and M.Tech., Aerospace Engineering — 2014 to 2019
Indian Institute of Technology (IIT Bombay)
Specialization in Systems and Control; Minor in Physics
Thesis Advisor: Prof. Srikant Sukumar

Research Interests

I develop theoretically grounded, decentralized decision and control frameworks for resource-constrained robotic systems. My research focuses on enabling robots to make fast, reliable, and provably safe decisions under limits on sensing, computation, communication, and actuation. I combine tools from nonlinear opinion dynamics, control theory, formal methods, and reinforcement learning to design scalable and adaptive autonomy for dynamic, uncertain, and infrastructure-limited environments.

Publications

*Advised students are marked with † and equal contributions with *, when applicable.*

Preprints and Manuscripts in Preparation

[P1] **Himani Sinhmar**, Vaibhav Srivastava, and Naomi Ehrich Leonard. "Neuromorphic Realization of Best Response in Finite-Action Games." arXiv preprint, 2026.

[P2] Ian Xul Belaustegui, **Himani Sinhmar**, Ling-Wei Kong, Andrew Michael Hein, and Naomi Ehrich Leonard. "A Continuous-Time and State-Space Relaxation of the Linear Threshold Model with Nonlinear Opinion Dynamics." arXiv preprint, 2026.

[P3] **Himani Sinhmar**, Vaibhav Srivastava, and Naomi Ehrich Leonard. "Neuromorphic Control for Provably Safe, Deadlock-Free, and Scalable Coordination of Mobile Robots in Crowded Spaces." In preparation.

[P4] Antoun K. Salib†, **Himani Sinhmar**, Vaibhav Srivastava, and Naomi Ehrich Leonard. "Real-Time Inference of Cooperation for Trajectory-Preserving Multi-Robot Social Navigation in Crowded Spaces." In preparation.

[P5] **Himani Sinhmar**, Giovanna Amorim†, Vaibhav Srivastava, and Naomi Ehrich Leonard. "Decentralized Neuromorphic Supervisory Control for Parsimonious Multi-Agent Persistent Environment Monitoring." In preparation.

Journal Articles

[J1] Marcus Greiff, **Himani Sinhmar**, Andreas Weiss, Kristofer Berntorp, and Stefano Di Cairano. "Invariant Set Planning for Quadrotors: Design, Analysis, and Experiments." *IEEE Transactions on Control Systems Technology*, 33(2):449-462, 2024.

[J2] Qingkun Liu*, Wei Wang*, **Himani Sinhmar**, Itay Griniasty, Jason Z. Kim, Jacob T. Pelster, Parag Chaudhari, Michael F. Reynolds, Michael C. Cao, David A. Muller, Alyssa B. Apsel, Nicholas L. Abbott, Hadas Kress-Gazit, Paul L. McEuen, and Itai Cohen. "Microscopic, Continuum, Compliant, and Electronically Configurable Metasheet Robots." *Nature Materials*, 2024.

[J3] **Himani Sinhmar** and Hadas Kress-Gazit. "Guaranteed Encapsulation of Targets with Unknown Motion by a Minimalist Robotic Swarm." *IEEE Transactions on Robotics*, 40:816-830, 2023.

[J4] Pallavi Sinha, Srikant Sukumar, and **Himani Sinhmar**. "Consensus of Networked Double Integrator Systems under Sensor Bias." *International Journal of Adaptive Control and Signal Processing*, 37(1):278-297, 2022.

Peer-Reviewed Conference Proceedings

[C1] Donggeon David Oh, Justin Lidard, Haimin Hu, **Himani Sinhmar**, Elle Lazarski, Deepak Gopinath, Emily S. Sumner,

Jonathan A. DeCastro, Guy Rosman, Naomi Ehrich Leonard, and Jaime Fernández Fisac. "Safety with Agency: Human-Centered Safety Filter with Application to AI-Assisted Motorsports." *Robotics: Science and Systems*, 2025.

[C2] **Himani Sinhmar** and Hadas Kress-Gazit. "Multi-Source Encapsulation With Guaranteed Convergence Using Minimalist Robots." *Distributed Autonomous Robotic Systems*, 2024.

[C3] **Himani Sinhmar**, Marcus Greiff, and Stefano Di Cairano. "Practical and Safe Navigation Function Based Motion Planning of UAVs." *IEEE International Conference on Robotics and Automation*, 2024.

[C4] **Himani Sinhmar** and Hadas Kress-Gazit. "Decentralized Control of Minimalistic Robotic Swarms for Guaranteed Target Encapsulation." *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.

[C5] **Himani Sinhmar** and Srikant Sukumar. "Distributed Model Independent Algorithm for Spacecraft Synchronization under Relative Measurement Bias." *5th CEAS Conference on Guidance, Navigation and Control*, 2019.

[C6] **Himani Sinhmar** and Vinod Kumar. "Relative Autonomous Navigation Without Communication Between Spacecraft Using Line of Sight Measurements." *8th IEEE/CSAA Guidance, Navigation and Control Conference*, 2018.

Workshop Papers and Abstracts

[W1] **Himani Sinhmar**, Itay Griniasty, Qingkun Liu, Wei Wang, Itai Cohen, and Hadas Kress-Gazit. "Design and Control of Microscopic Robot Sheet." *ICRA Workshop*, 2024.

Patents and Dissertation

[PT1] Marcus Greiff, Stefano Di Cairano, and **Himani Sinhmar**. "Motion Planning and Control with Multi-Stage Construction of Invariant Sets." US Patent Application No. 18/589,946, 2025.

[D1] **Himani Sinhmar**. "Control With Guarantees for Minimalist Robotic Swarms." Ph.D. dissertation, Cornell University, 2024.

Selected Honors and Awards

2026	**Future Values Fellowship**, Princeton University
2026	**Robotics Rising Star**, Princeton University
2025	**Future Leader in Robotics and AI**, University of Maryland
2024	**Hibbitt Postdoctoral Fellowship**, Brown University, offered and declined
2024	**Caltech EAS Trailblazers Award**, Caltech
2024	**Robotics: Science and Systems Pioneers**, RSS Conference
2022 to 2024	**Graduate Resident Fellowship**, Cornell University
2018	**Institute Undergraduate Research Award**, IIT Bombay

Presentations and Posters

Invited Seminars and Talks

2025	Invited participant, NSF visioning workshop "Formalization of Human Factors into Computational Models," sponsored by the Mind, Machine, and Motor Nexus program of NSF.	Alexandria, VA
2025	Invited seminar, Future leaders in robotics and AI, "Provably Scalable and Decentralized Control Design for Minimalist Robot Swarm" [Recording]	University of Maryland
2025	Invited Trailblazers seminar for Department of Computing + Mathematical Sciences	Caltech
May 2024	Leonard Lab, *Princeton University*.	Princeton, NJ
Feb. 2024	GRASP Spring Seminar, *University of Pennsylvania*.	Philadelphia, PA
Feb. 2024	Automatic Coordination of Teams Lab, *Brown University*.	Providence, RI
Jan. 2024	Biorobotics Lab, *Carnegie Mellon University*.	Pittsburgh, PA
Jan. 2024	Distributed Robotics Lab, *MIT CSAIL*.	Cambridge, MA
Jan. 2023	Collective Embodied Intelligence Lab, *Cornell University*.	Ithaca, NY
May 2022	Guest lecture, Introduction to Computational Science and Engineering, Cornell University.	Ithaca, NY

Conference and Poster Presentations

2026	Invited poster, Diverse Intelligences Summit.	Princeton University
2025	Poster, *Northeast Robotics Colloquium*.	Cornell University
2024	Participated in IEEE RAS Summer School on Multi-Robot Systems	Prague
2024	Poster, *Robotics, Science and Systems Conference (RSS)*.	TU Delft, Netherlands
2023	Poster, *International Conference on Robotics and Automation (ICRA)*.	Yokohama, Japan
2023	Poster, *Northeast Robotics Colloquium*.	Yale University

Teaching Experience

2021 to 2023	Autonomous Mobile Robots, Teaching Assistant, Cornell University.
2019	Dynamics and Control, Teaching Assistant, IIT Bombay.
2018	Spaceflight Mechanics, Teaching Assistant, IIT Bombay.

Mentoring and Professional Development

2026	Antoun K. Salib, Senior Thesis, Department of Electrical and Computer Engineering, Princeton University. Thesis: *Trajectory-Preserving Multi-Robot Navigation in Crowded Social Spaces with Online Cooperation Inference*.
2025	Facilitating Mentor Training, CIMER Academic Mentor Training Workshop.

Professional Service

Conference and Symposium Organization

2026	Organizing Committee, Princeton Robotics Symposium.
2025	Website and Publicity Chair, Robotics: Science and Systems Pioneers.
2022	Session Chair, Swarm Robotics, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
2018	Session Chair, Navigation Technology, IEEE/CSAA Guidance, Navigation and Control Conference.
2016 to 2017	Chief Editor, Department Newsletter *Lift-Off*, IIT Bombay.

Peer Review

- **Journals:** *Science Advances*; *IEEE Transactions on Robotics*; *IEEE Transactions on Automatic Control*; *The International Journal of Robotics Research*; *Autonomous Robots*; *IEEE Robotics and Automation Letters*.
- **Conferences:** Robotics: Science and Systems; IEEE Conference on Decision and Control; IEEE International Conference on Robotics and Automation; IEEE/RSJ International Conference on Intelligent Robots and Systems; International Symposium on Distributed Autonomous Robotic Systems.